

PADLOCK SYSTEM V1.6
DOCUMENTATION

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Introduction

Thank you for purchasing the “**Combination / Padlock**” system, this is a Raycast based system where the aim is to interact with a padlock and cycle through the combination spinners to find the correct combination to unlock it. Each of the combination spinners has their own number that you can set, and the “PadlockController” script controls the combination you can use to unlock it. It’s very easy to setup and even allows for more than one padlock per scene.



The asset includes:

- Raycast system that detects the padlock to bring up the large interact-able version in front of the camera.
- Add as many padlocks as are desired for your game.
- 4 spinner combination system with a 4 code string.
- PBR Padlock with brass / dark metal textures included.
- Animated latch to show the combination has been entered correctly.
- PBR door with basic animation and open script provided.
- Scripts which can be easily modified.
- Sound effects for the system are included

PLEASE VIEW THE ONLINE DOCUMENTATION HERE IF YOU REQUIRE MORE UPDATED HELP:

<https://speedtutoruk.gitbook.io/padlock-system-doc/>

FAQ

Q). How do I Import the asset?

A). Go to the Unity asset store and visit your “**My Assets**”. Download the asset if not already downloaded and click “**Import**”, import all required features of the asset for your use. It should have appeared in your project under “**Combination / Padlock System**”. (If Unity warns you that you may overwrite project data, do not choose the “**Project Settings**” tickbox when importing the asset)

Q). Is there an example of this asset working?

A). Yes, you can open the “**PadlockDemo**” to see the padlock asset in action or use this scene as your initial base of your project.

Q). How can I manually setup this asset?

A). See the manual setup instructions on [“Page 2”](#).

Q). How can I use a different Character Controller with the padlock system?

A). Check the [“Extending the padlock system”](#).

Q). How can I add more than one padlock set to my scene?

A). Check the [“Extending the padlock system”](#).

Q). How do I change the material of the padlock?

A). Check the [“Extending the padlock system”](#).

Q). Can I use this asset with a 3rd person character with this asset?

A). Yes, all you need is to use the trigger event and add the “**Padlockrigger**”. See additional note for details!

Q). My Raycast isn't working properly?

A). Make sure your player character (If it has a collider) uses a layer of “Ignore Raycast” to help this!

Manual Setup – Initial #1

PLEASE SEE THE ONLINE DOCUMENTATION FOR UPTO DATE SETUP, VIDEOS AND MORE! IF YOU NEED MORE HELP: <https://speedtutoruk.gitbook.io/padlock-system-doc/>

NOTE QUICK START: You can add the Entire_Padlock_DemoScene prefab to your scene to get everything setup and ready to go without having to follow any extra steps.

- 1). Make sure to create a Layer in the top right of the inspector called **PadlockSpinner**
- 2). Open up the Padlock Spawnable Objects in the prefabs folder and set the spinners to this layer
- 3). Make sure your main camera has a **Physics Raycaster** script and set the **EventMask** to **PadlockSpinner** too
- 4). Add the **FPSController** prefab from the **Prefabs** folder to your scene. (You can drag this into the hierarchy or the main scene.) Please navigate to the scripts folder and add the **PadlockInteractor** script to your **MainCamera**. (If not already).
 - Set your **MainCamera** to the tag of **MainCamera**
 - Set the **FPSControllers** layer to **Ignore Raycast**
 - Make sure the **Main Camera** has a **Physics Raycaster** script with the layer of **PadlockSpinner** in the EventMask
- 5). Add **PadlockTrigger / Padlock Raycast** sets to your hierarchy or scene from the assets **Prefabs** folder. Add this asset wherever you want in your scene.
 - Make sure your **Padlock_Interactable** GameObject has a:
 - **Box Collider**
 - **PadlockItem** script should already be on the prefab. If not, add it from the “Scripts” folder.
- 6). Open the **Padlock Spawnable** objects from the prefabs folder and each spinner should be setup as below:
 - Make sure to add the **PadlockNumberSelector** script – Set the row according to it's row
 - **Box Collider**
 - **Rigidbody**
 - Set the Layer to **PadlockSpinner**
- 7). Select **Padlock_Controller** object if not already added from the combined prefabs to your project, which will have the **PadlockController** script added

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8). Make sure to add the managers from the prefabs folder:

- **DisableManager** - Add the Player object and the main camera to these slots to find the appropriate reference
- **InputManager** - These are the inputs for the system, change as required
- **AudioManager** - Add all of the appropriate ScriptableObject audio effects from the scriptableobjects folder
- **UIManager** - This UI manager will have the canvas for all UI elements parented and you should add the correct UI elements to the slots in the manager.

Final Notes

REFERENCING THE AUDIO MANAGER:

1). If you click on the audio manager you can add addition elements to the array by incrementing the value by as many sound clips as you want to add, make sure to give them a name you will remember as you will reference this in your code!

2). Use the code, within your scripts to reference your sounds to play!

PLAudioManager.instance.Play(ScriptableObjectName);

ADDING A TRIGGER EVENT – Using PadlockTrigger script for 3rd person Controllers

1). Create a box or other 3D Object, make sure this object has a collider set to **IsTrigger**

2). Add the **PadlockTrigger** script to that 3D object and add the keypad object to the inspector slot named **PadlockController**

3). Then set an input key you'd like to use for the interaction, by default set to "E"!

Remember to take a look at the demo scene if you have any troubles, it might give you an idea on how to fix an issue!

If you find the package helpful, please leave a positive review and star rating as it would really help me out! 😊 If you have any problems, feel free to send an email to me!

Extending the Padlock System

I want to do something different when the code is valid:

In the padlock controller script object, this uses **UnityEvents** which allows you to click the + icon in the bottom right and then add your own objects with a script with a public method. I have an example in the demo scene, which uses the door object with the **DoorAnimation** script.

How to add another padlock set to the same scene?

Duplicate the **Padlock_Interactive** and **Padlock_Controller** from your scene and increment the numbers like I mentioned in the Final Notes section.

I want to change the material on the padlock to the alternative, how do I do this?

Visit the **Materials** folder of the package and look for the folder named **M_Padlock_Black** and open the folder. Add the **M_Padlock_Spinner_Black** to each of your spinners and the **M_Padlock_Main_Black** to the main body and metal latch of the interactable object and mechanism that resides on your camera or add the prefabs to your scene!

How do I use another controller with this system?

In the **DisableManager** script we reference the player controller so we can disable the **FirstPersonController** script so when we interact with the padlock we disable character movement and camera rotation. If you're using a different controller please replace these lines with appropriate for you.

```
namespace PadlockSystem
{
    # Unity Script | 4 references
    public class PLDisableManager : MonoBehaviour
    {
        [SerializeField] private FirstPersonController player = null;
        [SerializeField] private PadlockInteractor mainCameraInteractor = null;

        public static PLDisableManager instance;

        # Unity Message | 0 references
        void Awake()
        {
            if (instance != null) { Destroy(gameObject); }
            else { instance = this; DontDestroyOnLoad(gameObject); }
        }

        3 references
        public void DisablePlayer(bool disable)
        {
            if (disable)
            {
                player.enabled = false;
                mainCameraInteractor.enabled = false;
                PLUIManager.instance.DisableCrosshair(true);
            }

            else
            {
                player.enabled = true;
                mainCameraInteractor.enabled = true;
                PLUIManager.instance.DisableCrosshair(false);
            }
        }
    }
}
```

Online Documentation

<https://speedtutoruk.gitbook.io/padlock-system-doc/>

Contact

If you have any problems with the pack, or have some ideas for new features you'd be interested in, please feel free to contact me.

Thank you very much for downloading! Please be sure to leave a **5*** rating and **review** if you liked the package! Support me on

Patreon: <https://www.patreon.com/SpeedTutor>

Make a small donation: <http://www.paypal.me/speedtutor>

If you have any problems about the setup, usage, customisation or have any suggestions for future updates of this kit, please feel free to contact me.

Email: speedtutoruk@gmail.com

Discord: <https://discord.gg/FguwaJHfSY>

YouTube: <https://www.youtube.com/user/speedtutor>

Website: <http://www.speed-tutor.com>

Patch Notes

Version 1.6 – June 2024

- **General fixes and brand new null ref check solutions**

Version 1.5 - July 2023

- Updated the brass texture to make sure numbers are fully visible with an option of white texture texture
- Fixed script GUID issues by duplicating all scripts to avoid script conflicts
- Made **Sound** script generic
- Renamed UI elements and inverted trigger interact prompt colours for easier visuals
- **PadlockController** script
 - Renamed **PLController** to **PadlockController**
 - Refactored the entire script and remove redundant fields
 - Added brand new functionality to spawn padlocks instead of having them exist in the scene meaning it's very easy to setup and use
- **PadlockNumberSelector** script
 - Removed the need for adding the PadlockController
 - Added IPointerEnter interaction to PadlockSpinners for better interaction
 - Added Physics Raycaster to make this into account
- **PadlockInteractor** script
 - Renamed the script from the **PLRaycast** from **PadlockInteractor**
 - Organised the script
- **AudioManager** script
 - Added ScriptableObject update to the Play and Stop methods for easier functionality
- **PadlockTrigger** script
 - Organised this script and made it more split between methods

Version 1.4 - June 2022

- Organised project files and optimised texture naming
- Organised demo scene for ease of use
- Created a new UI manager to control all UI elements in one place
 - Added crosshair UI control to the UI manager from the raycast script
 - Added basic disabling functionality to the UI manager
- Created ScriptableObjects for all audioclips
- Edited the disable manager to be more concise with new UI manager

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Version 1.3 - June 2021

- Updated the "PadlockRaycast" script
 - o Added "Ignore Raycast" layer to the to player
 - o No need for multiple tags within the system
- Added a material and shader to allow 3D text to be visible from one side only, removing the need for several other materials
- Removed redundant materials and textures from the projects to save space
- Removed light baking data from the demo scene to save space
- Added the Raycast disabling reference to the disable manager for consistency and ease of access
- Added a variable reference to the "DoorAnimation" script to change animation name easily
- Refactored "PadlockController" script
 - o Removed the need for multiple references to lock animations and camera, these are found automatically
 - o Added audio clip sounds to the inspector to make customisation easier
 - o Renamed variables to make them easier to understand and more consistent

Version 1.2 – July 2020

- UI Changes:
 - o Added a button prompt when using a "Trigger Style" event for 3rd person style interaction
- Scripting Changes
 - o Updated the Padlock Controller to incorporate "**UnityEvents**" to make successful code interaction multi-use, meaning you can produce multiple outcomes with ease
 - o Added a new Audio Manager, so more sounds can be added and called within script
 - o Added a new input manager, which allows easy control of all inputs from one place
 - o Refactored the Raycast, PadlockController, ScriptSpinners and more!
- Miscellaneous
 - o Updated and edited sounds, textures and materials
 - o Added new manager examples to the prefabs and demo scene
 - o Update all prefabs and set them to 0, 0, 0 for future use
 - o Added an additional Trigger Style interaction with 3rd person or possible first person instances

Version 1.1.1 – May 2020

- General Fixes
 - o Added namespaces to scripts to stop any clashes with other projects
 - o Added a new LayerMask called "LayerToExclude" which will allow you to add a layer to a wall or door (For example) and stop the raycast going through.
 - o Updated demo scene FPSPrefab and general prefabs to be more consistent